

A difference-system model for condition (A), and the constant-defect case

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Abstract

Condition (A) — log-concavity in a of the cylindric Kostka number $k(a, b)$ at fixed b — is re-expressed exactly as the lattice-point count

$$k(a, b) = \#\{(y, r) \in \mathbb{Z}^m \times \mathbb{Z}^m : y \in B, \sum_i y_i = \sigma, 0 \leq r_i \leq g_i, y_i \leq r_i, r_i \leq g_i + y_{i+1}, \sum_i r_i = a + \sigma\},$$

in which *every* constraint bounds a single coordinate or a difference of two coordinates (Theorem 3). So the ambient set is the lattice-point set of an alcoved polytope, closed under componentwise $\wedge$ and $\vee$, and (A) asks that its fibre counts along one of two coordinate-sum functionals be log-concave. In these coordinates the two proved structural results of the programme — concentricity (**A-general-m-concentricity**) and the ℓ^1 half-width formula (**A-half-width-is-l1-distance**) — both collapse to the single scalar identity $(t)_+ - (-t)_+ = t$ summed over the hyperplane (Corollary 4).

Two consequences are proved. First, the *support shift* of each summand equals the horizontal-strip defect, so on a slice of *constant* defect — equivalently a single half-width, equivalently (by Theorem R of [3]) an M -convex width-vector set — the shift is constant; and if in addition every $y_i \geq 0$ on the slice, or every $y_i \leq 0$, the width map is *affine* and the width-vector set is a *box slice*. Condition (A) on such a slice is then equivalent to

$$(Q) \quad F(z) = \sum_{w \in \text{Box} \cap \{\sum w_i = T\}} \prod_{i=1}^m [w_i]_z \in \text{PF}_2, \quad [w]_z = 1 + z + \cdots + z^{w-1}.$$

Second, (Q) is proved at $m = 2$ — by *concavity*, which is strictly stronger than log-concavity — and the mechanism is shown to fail at $m \geq 3$ for an exactly identified reason. (Q) is restated as a count over an M -convex set $\Omega \subset \mathbb{Z}^{2m}$ cut out by a laminar family (Theorem 12), its support is shown to be an interval for free, and the version with the second hyperplane *removed* is proved PF_2 outright from **pf2-convolution** (Theorem 16) — which locates the whole difficulty in that one constraint. Measured: (Q) holds on 20 471 exhaustive and 60 000 random instances ($m \leq 6$, $|W|$ up to 32 765, degree up to 59), 0 failures.

The brief's twenty-minute first move is reported as a **vacuous pass**, the third instance of that shape: neither witness of **A-shape-blind-impossible** is realisable in the class the briefed hypothesis (H3) carves out (§1), so the discriminator excluded both sides. Two routes to (Q) at $m \geq 3$ are located and both are blocked, with the obstruction named in each case (§7).

1 What was asked, and what happened

The brief named one hypothesis and one twenty-minute discriminator.

Hypothesis 1 ((H3), as briefed). Does M -convexity of the whole *nest* $\{\Sigma_b^{\leq j}\}_{j \geq 0}$ (Theorem M of [3]), together with the explicit width map $w : \nu \mapsto (w_1(\nu), \dots, w_m(\nu))$, imply that the slice sum $k(\cdot, b)$ is PF_2 ?

The discriminator: compute the nest for each of the two multisets of **A-shape-blind-impossible** (**thm:blind** of [1]),

$$\begin{aligned} A &= \{(1, 1), (2, 2), (1, 5), (2, 6)\} \longrightarrow (1, 3, 4, 6, 4, 3, 1) \text{ not } \text{PF}_2, \\ B &= \{(1, 1), (2, 2), (3, 3), (4, 4)\} \longrightarrow (1, 3, 6, 10, 6, 3, 1) \text{ } \text{PF}_2, \end{aligned}$$

both with half-width profile $M = (1, 1, 1, 1)$. If the nests coincide, (H3) is shape-blind and dies at once; if they differ, (H3) is the first candidate to survive **thm:blind**.

Prediction, written before the run. The nests differ, because the level- j width multisets differ.

What the run returned. Both sums and both PF_2 verdicts were reproduced exactly by the instrument before anything else was computed. Since $k(\nu) = c - \Lambda(\nu)$ (eq. (2) below), the nest is the filtration by half-width, so with $c = 3$:

$$\begin{array}{cccc} j = 0 & j = 1 & j = 2 & j = 3 \\ A : & \{(2, 6)\} & \{(1, 5), (2, 6)\} & \{(1, 5), (2, 2), (2, 6)\} \quad \text{all four} \\ B : & \{(4, 4)\} & \{(3, 3), (4, 4)\} & \{(2, 2), (3, 3), (4, 4)\} \quad \text{all four} \end{array}$$

with $|\Sigma^{\leq j}| = 1, 2, 3, 4$ for *both*. The predicted answer, hence the weaker one.

The question the brief did not ask, and the answer that matters. Is either witness realisable *at all* in the class (H3) carves out — i.e. as $\{w(y) : y \in B \cap \{\sum y = \sigma\}\}$ for some box B , some σ and some gap vector g ? No, and the reason is a one-line identity.

Proposition 2 (the first move is a vacuous pass). *At $m = 2$, on any slice $Y = B \cap \{y_1 + y_2 = \sigma\}$,*

$$w_1(y) - w_2(y) = (g_1 - g_2) - y_1 + y_2 = (g_1 - g_2) + \sigma - 2y_1,$$

so $y \mapsto w_1(y) - w_2(y)$ is injective on Y . Hence the width differences of any realisable multiset are pairwise distinct. Witness A has differences $(0, 0, -4, -4)$ and witness B has $(0, 0, 0, 0)$; neither is realisable. Exhaustively: over all 43 008 valid four-point abstract $m = 2$ slices with $g_i \leq 16$ and both σ and the run-start in $[-32, 32]$, there are 0 realisations of A and 0 of B .

Proof. From eq. (1), $w_1 - w_2 = (g_1 - g_2) - (y_1)_+ + (-y_1)_+ + (y_2)_+ - (-y_2)_+$, and $(t)_+ - (-t)_+ = t$ gives the displayed form. It is affine in y_1 with slope -2 , and $y_2 = \sigma - y_1$ is determined by y_1 , so $w_1 - w_2$ determines y . $\square$

So (H3) excludes *both* sides of the intended discriminator: the test could not have come out any other way, and it is a non-refutation, not evidence. This is the third consecutive session in which the briefed first move has been vacuous — 10-03's **prop:firstmove** (the non- PF_2 certificate was not M -convex, but neither was the PF_2 one) and 10-04 c1's **A-level-wise-conditions-insufficient** record the same shape one level down. The generator is stable: a discriminator built from two objects that are known to differ *in the conclusion* says nothing unless both objects satisfy the *hypothesis*, and nothing in my instruments grades that. The repair that worked here is cheap and

should be standing: *before running a discriminator, check that each of its two inputs lies in the class the hypothesis quantifies over.*

The rest of this note abandons the nest as a hypothesis and uses it as what it actually is — a filtration by half-width whose *bottom and top* levels are where the width map becomes affine.

2 Setup

Throughout $m \geq 1$, $n \geq m$, $G := n - m$, and $\mu \leq \lambda$ are cylindric shapes in $C^{n,m}$ (bi-infinite strictly increasing with $x_{i+m} = x_i + n$, stored as $(x_1, \dots, x_m)$, indices cyclic, $\lambda_0 = \lambda_m - n$). Write $d = |\lambda/\mu|$. We use the following premises from `proofs/registry/cylindric-lorentzian.json`, with their stored trust levels.

- **A-slice-sum-reformulation** (*proved*): with $L_i(\nu) = \max(\nu_i, \lambda_{i-1} + 1)$, $R_i(\nu) = \min(\nu_{i+1} - 1, \lambda_i)$, $w_i = R_i - L_i + 1$, $\text{Trap}_\nu = *_i \mathbf{1}_{[0, w_i - 1]}$ shifted by $\sum_i (L_i - \nu_i)$,

$$k(a, b) = \sum_{\nu \in \Sigma_b} \text{Trap}_\nu(a), \quad \Sigma_b = \{\nu \in \text{Box}(\mu) : S_\nu = |\mu| + b\}, \quad \text{Box}(\mu) = \prod_i [\mu_i, \mu_{i+1} - 1].$$

- **A-half-width-is-l1-distance** (*lean-verified*): with $a_i = \lambda_{i-1} + 1$, $y_i = \nu_i - a_i$, $g_i = \lambda_i - \lambda_{i-1} - 1 \geq 0$ (so $\sum_i g_i = G$) and $\sigma = \sum_i y_i$,

$$w_i = g_i + 1 - (y_i)_+ - (-y_{i+1})_+, \quad (1)$$

$$\Lambda(\nu) = \frac{1}{2}(G - \|y\|_1), \quad c - \Lambda(\nu) = k(\nu) := \sum_i (-y_i)_+ \geq 0, \quad (2)$$

with $k(\nu) = 0$ iff λ/ν is a cylindric horizontal strip, and $c = (d - b)/2 = (G - \sigma)/2$ the common centre.

- **pf2-convolution** (*proved*; a *paper* proof via Toeplitz matrices and Cauchy–Binet, *not* formalised): $\text{PF}_2 = \log\text{-concave}$ with interval support is closed under convolution.
- **raw-hessian-lemma** (*proved*): the Hessian of $\partial^\beta N(h)$ has entries $M_{ij} = c_{\beta+e_i+e_j}$, the *raw* coefficients.
- **13-det-reduction** (*lean-verified*): a symmetric 3×3 matrix with nonnegative entries has at most one positive eigenvalue iff $e_2(M) \leq 0$ and $\det M \geq 0$.
- **A-shape-blind-impossible** (*proved*): `thm:blind`, no hypothesis in the half-widths alone can suffice.
- Theorem M and Theorem R of [3] (*proved*): every $\{y \in B : \sum y_i = \sigma, \sum_i (-y_i)_+ \leq j\}$ is M -convex; and M -convexity of the width-vector set forces a single half-width.

3 The difference-system model

Theorem 3 (difference-system model for $k(a, b)$). *Put $y_i = \nu_i - a_i$ and $r_i = \kappa_i - a_i$, where $a_i = \lambda_{i-1} + 1$, and let $B = \prod_i [\mu_i - a_i, \mu_{i+1} - 1 - a_i]$, $\sigma = |\mu| + b - \sum_i a_i$. Then*

$$k(a, b) = \# \left\{ (y, r) \in \mathbb{Z}^m \times \mathbb{Z}^m : \begin{array}{l} y \in B, \quad \sum_i y_i = \sigma, \\ 0 \leq r_i \leq g_i, \quad y_i \leq r_i, \quad r_i - y_{i+1} \leq g_i, \\ \sum_i r_i = a + \sigma \end{array} \right\}. \quad (3)$$

Every constraint in (3) other than the two coordinate-sum equalities bounds a single coordinate or a difference of two coordinates. Consequently the ambient set

$$\Gamma = \{(y, r) : y \in B, 0 \leq r_i \leq g_i, y_i - r_i \leq 0, r_i - y_{i+1} \leq g_i\}$$

is the lattice-point set of an alcoved polytope and is closed under componentwise $\wedge$ and $\vee$.

Proof. $k(a, b)$ counts chains $\mu \subseteq \nu \subseteq \kappa \subseteq \lambda$ of cylindric horizontal strips with $|\nu/\mu| = b$ and $|\kappa/\nu| = a$. Translate each of the three strip conditions into bead coordinates. Since $a_{i+1} - a_i = \lambda_i - \lambda_{i-1} = g_i + 1$:

- $\mu \subseteq \nu$ a horizontal strip $\iff \mu_i \leq \nu_i \leq \mu_{i+1} - 1$, i.e. $y \in B$; and $|\nu/\mu| = b \iff S_\nu = |\mu| + b \iff \sum_i y_i = \sigma$.
- $\kappa \subseteq \lambda$ a horizontal strip $\iff \lambda_{i-1} < \kappa_i \leq \lambda_i \iff 0 \leq r_i \leq g_i$.
- $\nu \subseteq \kappa$ a horizontal strip $\iff \nu_i \leq \kappa_i < \nu_{i+1}$. The left half is $y_i \leq r_i$. For the right half, $\nu_{i+1} = a_{i+1} + y_{i+1} = a_i + g_i + 1 + y_{i+1}$, so $\kappa_i < \nu_{i+1} \iff r_i \leq g_i + y_{i+1}$.
- $a = |\kappa/\nu| = \sum_i (\kappa_i - \nu_i) = \sum_i (r_i - y_i) = \sum_i r_i - \sigma$.

Cylindricity of κ is automatic: $\kappa_i \leq \lambda_i < \lambda_{i+1} < \kappa_{i+2} + n$ is not needed — $\kappa_i < \nu_{i+1} \leq \kappa_{i+1}$ already gives strict increase, and periodicity is inherited from λ . The last assertion is the standard fact that a system of difference constraints $x_p - x_q \leq c_{pq}$ (with single-coordinate bounds the special case $q = \text{ground}$) defines a set closed under $\wedge$ and $\vee$: if $x_p - x_q \leq c$ and $x'_p - x'_q \leq c$ and $\max(x_p, x'_p) = x_p$, then $\max(x_p, x'_p) - \max(x_q, x'_q) \leq x_p - x_q \leq c$, and dually for min. $\square$

Corollary 4 (concentricity and the half-width formula, for free). *For $y \in B$ with all widths positive, the r -interval at bead i is $[(y_i)_+, g_i - (-y_{i+1})_+]$, of length w_i as in (1). Hence*

$$\sum_i (\text{lower}_i + \text{upper}_i) = \sum_i ((y_i)_+ + g_i - (-y_{i+1})_+) = G + \sum_i ((y_i)_+ - (-y_i)_+) = G + \sigma,$$

which is constant on the slice: this is **A-general-m-concentricity**. And

$$\sum_i w_i = G + m - \sum_i ((y_i)_+ + (-y_i)_+) = G + m - \|y\|_1,$$

which is (2). Both are the single identity $(t)_+ - (-t)_+ = t$ (resp. $(t)_+ + (-t)_+ = |t|$) summed over the hyperplane.

Remark 5 (the shift is the defect). $L_i - \nu_i = \max(0, \lambda_{i-1} + 1 - \nu_i) = (-y_i)_+$, so the support shift of Trap_ν is $\sum_i (-y_i)_+ = k(\nu)$, the horizontal-strip defect. Together with $c = k(\nu) + \Lambda(\nu)$ this is why the summands are concentric. In particular the shift is constant on a slice exactly when the defect is, i.e. exactly when there is a single half-width.

Remark 6 (calibration). (3) reproduces $k(\cdot, b)$ exactly on all 155 402 nonempty slices with $2 \leq m \leq 5, m \leq n \leq m+4, d \leq 7$, checked against the box-slice formula of **A-slice-sum-reformulation**, and on all 2491 slices checked against **gen.k.table.chains**, a direct chain enumeration that uses neither L_i, R_i nor y, r . Two refusal controls bite: replacing $r_i \leq g_i + y_{i+1}$ by $r_i \leq g_i + y_i$ (decoupling the cycle) disagrees on 1231 of 2493, and by $r_i \leq g_i + y_{i+1} + 1$ on 1956 of 3065.

The dual form, and a route it kills

Summing over y first instead of r gives

$$k(a, b) = \sum_{\substack{r \in \prod_i [0, g_i] \\ \sum_i r_i = a + \sigma}} N(r), \quad N(r) = \#\left\{y : \max(P_i, r_{i-1} - g_{i-1}) \leq y_i \leq \min(Q_i, r_i), \sum_i y_i = \sigma\right\}, \quad (4)$$

so $k(\cdot, b)$ is the *slice-sum transform of one fixed nonnegative function N on a box* — the a -dependence has moved entirely into which slice is being summed. This is the cleanest form of (A) I have, and it immediately suggests the hypothesis that N is $M^\natural$ -concave, since the slice-sum transform of a product of PF_2 factors is PF_2 by **pf2-convolution**. That hypothesis is **false**: (4) agrees with (3) on all 14 021 slices tested, and among the 1842 with $|\text{supp } N| \geq 2$ there are 8 on which N is not $M^\natural$ -concave, the smallest at $m = 4$, $n = 7$, $\mu = (0, 1, 3, 5)$, $\lambda = (1, 2, 4, 6)$, $b = 1$: $g = (1, 0, 1, 1)$ and $N = \{0000:1, 0001:2, 0010:2, 0011:3, 1000:1, 1001:1, 1010:1, 1011:1\}$, whose slice sums $(1, 5, 5, 1)$ are nevertheless PF_2 . So the dual discrete-concavity route is closed.

4 Conjecture N: the abstract-box generalisation

Theorem M of [3] holds for *every* box. So every triple (g, B, σ) satisfies (H3)'s hypotheses, and the class (H3) carves out is

$$\mathcal{C} = \left\{ (g, B, \sigma) : g \in \mathbb{Z}_{\geq 0}^m, B = \prod_i [P_i, Q_i], Y = B \cap \{\sum y_i = \sigma\} \neq \emptyset, w_i(y) \geq 1 \forall y \in Y, \forall i \right\},$$

which contains every real slice.

Question 7 (Conjecture N). Is $\sum_{y \in Y} \text{Trap}_{w(y)}$ (summands centred by Remark 5) in PF_2 for every $(g, B, \sigma) \in \mathcal{C}$?

Measured: 134 944 instances at $m = 3, 4, 5$ ($G = \sum g_i$ up to 69, $|Y|$ up to 1978) and 675 892 exhaustive at $m = 2$ ($g_i \leq 6$, box in $[-6, 6]$): 0 failures.

Controls, and which of them were controls.

- **C1, arbitrary subset of the box slice: REFUSES.** Replacing Y by a random proper subset fails on 20/6000 at $m = 3$ and 4/6000 at $m = 4$ (witness: $m = 3$, $g = (8, 13, 0)$, $Y = \{(5, 6, -3), (7, 5, -4), (3, 5, 0), (6, 6, -4)\}$, widths $\{(2, 5, 1), (3, 4, 1), (4, 5, 1), (6, 9, 1)\}$, sum $(1, 2, 3, 5, 9, 13, 15, 15, 13, 9, 5, 3, 2, 1)$, which fails PF_2 first at $3^2 = 9 < 2 \cdot 5 = 10$ — on the shoulder, not at the centre). So the *box-slice* structure of Y is load-bearing, and Question 7 is not vacuous.
- **C2–C5 were not controls, and this is the point.** Perturbing eq. (1) by pairing i with $i + 2$, or by taking $s_i = (y_i)_+ + (y_{i+1})_+$, or $s_i = (-y_i)_+ + (-y_{i+1})_+$, refuses 0/6000 in every case. The reason is Corollary 4: each of these perturbations still has $\sum_i (\text{lower}_i + \text{upper}_i)$ constant on the slice (equal to $G + \sigma$, G , G respectively), i.e. each is still *concentric*, hence each is another instance of the same statement rather than a perturbation of it. A control that never refuses is not a control; here the reason it never refuses is that Question 7 is more general than the hypothesis it was written for, and the correct generality is: *intervals* $[\alpha_i(y), \beta_i(y)]$ with α_i convex and 1-Lipschitz in a single coordinate, β_i concave and 1-Lipschitz in a single coordinate, and $\sum_i (\alpha_i + \beta_i)$ constant on Y .

- **C6, two hyperplanes** ($\sum y_i \in \{\sigma, \sigma + 1\}$): **NOT DECIDABLE** by this instrument. The first run reported 0/6000 refusals. That number is meaningless: mixed parity of $\sum_i w_i$ puts the summands on both parity classes of the half-integer grid, and `centred_sum` was silently keeping one class and dropping the other. A guard now returns UNDEFINED in that case. Recorded rather than deleted, because the failure mode is a silent near-zero of exactly the kind that reads as a pass.

5 The constant-defect case, and the statement (Q)

Theorem 8 (affine width map at the two ends of the nest). *Let Σ_b° be an effective slice.*

1. *If $y_i \geq 0$ for every y in the slice and every i — equivalently the horizontal-strip defect vanishes identically, i.e. $\Sigma_b^\circ = \Sigma_b^{\leq 0}$ — then $w_i = g_i + 1 - y_i$ is affine, the shift is 0, and*

$$W = \{w(y) : y \in \Sigma_b^\circ\} = \left\{w : L_i \leq w_i \leq U_i, \sum_i w_i = T\right\}, \quad L_i = g_i + 1 - Q_i, \quad U_i = g_i + 1 - P_i, \quad T = G + m - \sigma,$$

a box slice — in particular an M -convex set, with the single half-width $(T - m)/2$ demanded by Theorem R.

2. *If $y_i \leq 0$ for every y and every i , then with $z_i = -y_{i+1} \geq 0$ one has $w_i = g_i + 1 - z_i$, the shift is the constant $-\sigma$, and W is again a box slice.*

In both cases, writing $l_i = L_i - 1$, $h_i = U_i - 1$, $D = T - m$,

$$k(a + \text{shift}, b) = \sum_{w \in W} \text{Trap}_w(a) = \#\left\{(x, e) \in \mathbb{Z}_{\geq 0}^m \times \mathbb{Z}_{\geq 0}^m : l_i \leq x_i + e_i \leq h_i, \sum_i x_i = a, \sum_i e_i = D - a\right\}.$$

Hence condition (A) on such a slice is exactly the statement

$$(Q) \quad F(z) = \sum_{w \in \text{Box} \cap \{\sum_i w_i = T\}} \prod_{i=1}^m [w_i]_z \in \text{PF}_2.$$

Proof. (1) If $y_i \geq 0$ then $(y_i)_+ = y_i$ and $(-y_{i+1})_+ = 0$, so (1) reads $w_i = g_i + 1 - y_i$; the map $y \mapsto w$ is the composition of $y \mapsto -y$ with a translation, a bijection of $\mathbb{Z}^m$ carrying the box $\prod [P_i, Q_i]$ onto $\prod [g_i + 1 - Q_i, g_i + 1 - P_i]$ and the hyperplane $\sum y_i = \sigma$ onto $\sum w_i = G + m - \sigma$. The shift is $\sum_i (-y_i)_+ = 0$ by Remark 5. (2) If $y_i \leq 0$ then $(y_i)_+ = 0$ and $(-y_{i+1})_+ = -y_{i+1} = z_i$, so $w_i = g_i + 1 - z_i$, and z ranges over the cyclic reindexing of $-B$ intersected with $\sum z_i = -\sigma$, again a box slice; the shift is $\sum_i (-y_i) = -\sigma$. For the count: $\text{Trap}_w(a) = \#\{x : 0 \leq x_i \leq w_i - 1, \sum x_i = a\}$; put $e_i = w_i - 1 - x_i \geq 0$, so $x_i + e_i = w_i - 1 \in [l_i, h_i]$ and $\sum_i e_i = \sum_i (w_i - 1) - a = D - a$. The correspondence $(w, x) \leftrightarrow (x, e)$ is a bijection. $\square$

Proposition 9 (reach, measured). *Over all 25 768 nonempty effective slices with $2 \leq m \leq 5$, $m \leq n \leq m + 4$, $d \leq 7$ (of which 8619 have $|\Sigma_b^\circ| \geq 2$): the hypothesis $y \geq 0$ holds on 9203 slices (1652 nontrivial), $y \leq 0$ on 3909 (480), so Theorem 8 covers 12 665 slices (2132 nontrivial). A single half-width occurs on 20 322 slices (3173 nontrivial), and on exactly those 20 322 the width-vector set W is a box slice. So the hypothesis of Theorem 8 is strictly stronger than (Q)’s: 7657 slices (1041 nontrivial) have W a box slice without being one-sided, and (Q) covers those too.*

Remark 10. Proposition 9 is also the honest answer to the brief’s question “does Theorem M buy condition (A)?”. It does not buy (A); what it buys is the *indexing* of the nest, and the two *ends* of that nest — $j = 0$ and $j = \max$ — are precisely where the width map becomes affine. The information-carrying level is not an interior j ; it is the two extremes.

6 What is proved about (Q)

Proposition 11 (interval support, for free). *If the box slice W is nonempty then $\text{supp } F = [0, D]$ exactly. Hence (Q) is equivalent to log-concavity of the coefficients of F .*

Proof. Trap_w has support $[0, \sum_i (w_i - 1)] = [0, D]$ for every $w \in W$, since each factor $\mathbf{1}_{[0, w_i - 1]}$ has $w_i \geq 1$. A sum of nonnegative sequences with common support $[0, D]$ has support $[0, D]$. $\square$

Theorem 12 (Ω is M -convex). *Let $l_i \leq h_i$ be integers, $D \in \mathbb{Z}$, and*

$$\Omega = \left\{ (x, e) \in \mathbb{Z}_{\geq 0}^m \times \mathbb{Z}_{\geq 0}^m : l_i \leq x_i + e_i \leq h_i \ (1 \leq i \leq m), \sum_i x_i + \sum_i e_i = D \right\} \subset \mathbb{Z}^{2m}.$$

Then Ω satisfies the symmetric exchange axiom: for all $z, z' \in \Omega$ and every coordinate p with $z_p > z'_p$ there is a coordinate q with $z_q < z'_q$ such that $z - e_p + e_q \in \Omega$ and $z' + e_p - e_q \in \Omega$.

Proof. Write $z = (x, e)$, $z' = (x', e')$, $v_i = x_i + e_i$, $v'_i = x'_i + e'_i$; then $\sum_i v_i = \sum_i v'_i = D$. By the symmetry $x \leftrightarrow e$ we may assume p is the coordinate x_1 , so $x_1 > x'_1$.

Case (ii): $v_1 \leq v'_1$. Then $u := e_1$ satisfies $e_1 = v_1 - x_1 < v'_1 - x'_1 = e'_1$, so $q = e_1$ is admissible. Taking $q = e_1$ keeps the pair sum v_1 (resp. v'_1) unchanged, so the pair constraints hold; $x_1 - 1 \geq 0$ because $x_1 > x'_1 \geq 0$, and $e'_1 - 1 \geq 0$ because $e'_1 > e_1 \geq 0$. Both images lie in Ω .

Case (i): $v_1 > v'_1$. Since $\sum v = \sum v'$ there is j with $v_j < v'_j$, and $j \neq 1$. Within pair j , $v_j < v'_j$ forces $x_j < x'_j$ or $e_j < e'_j$; let q be a coordinate of pair j with $z_q < z'_q$. For $z - e_{x_1} + e_q$: $x_1 - 1 \geq 0$ as above, $z_q + 1 \geq 0$; pair 1 drops to $v_1 - 1 \geq l_1$ because $v_1 > v'_1 \geq l_1$; pair j rises to $v_j + 1 \leq h_j$ because $v_j < v'_j \leq h_j$. For $z' + e_{x_1} - e_q$: $z'_q - 1 \geq 0$ because $z'_q > z_q \geq 0$; pair 1 rises to $v'_1 + 1 \leq h_1$ because $v'_1 < v_1 \leq h_1$; pair j drops to $v'_j - 1 \geq l_j$ because $v'_j > v_j \geq l_j$. Both images lie in Ω . $\square$

Remark 13 (what Theorem 12 uses, and a control that was no control). The only hypothesis consumed is that the constraint family — singletons, the pairs $\{x_i, e_i\}$, and the whole ground set — is *laminar*. Verified: 27 816 sets Ω ($m = 2, 3, 4$, $l_i \leq 2$, $h_i \leq 3$, all D) are M -convex, 0 failures, $|\Omega|$ up to 1905 (sets above 120 points skipped: 10 989).

My first refusal control for this theorem **was no control**: I “dropped the pair structure” by replacing $l_i \leq x_i + e_i \leq h_i$ with $l_i \leq x_i \leq h_i$, which is a box intersected with a hyperplane in $\mathbb{Z}^{2m}$ — itself M -convex — so it returned 1037/1037 green and I briefly read that as a confirmation. The genuine control breaks *laminarity*: adding crossing bounds $c \leq e_i + x_{i+1} \leq c'$ (the family $\{x_i, e_i\}$, $\{e_i, x_{i+1}\}$ overlaps without nesting) refuses 517/897, 1817/1987 and 2128/2899 at three choices of (c, c') , while the laminar family itself refuses 0/3227 on the same enumerator.

Theorem 14 ((Q) at $m = 2$, by concavity). *For $m = 2$, F is concave on $[0, D]$ and strictly positive there; hence F is log-concave with interval support, i.e. (Q) holds at $m = 2$.*

Proof. For $m = 2$ and $w_1 + w_2 = T$ fixed, $\text{Trap}_{(w_1, w_2)}(a) = \min(a, D - a, w_1 - 1, w_2 - 1) + 1$ for $0 \leq a \leq D = T - 2$. Put $M(a) = \min(a, D - a)$ and $j(w) = \min(w_1 - 1, w_2 - 1)$. Then

$$F(a) = \sum_{w \in W} (\min(M(a), j(w)) + 1) = \sum_{j \geq 0} n_j (\min(M(a), j) + 1), \quad n_j = \#\{w \in W : j(w) = j\}.$$

M is concave on $[0, D]$ (a minimum of two affine functions), and $t \mapsto \min(t, j) + 1$ is concave and nondecreasing, so each summand is concave in a ; a nonnegative combination of concave functions is concave. Positivity on $[0, D]$ is Proposition 11. Finally log is concave and increasing, so $\log F$ is concave: F is log-concave. $\square$

Proposition 15 (the concavity mechanism stops at $m = 3$, and why). F is concave on the interior of its support on all 1456 exhaustive cases at $m = 2$ ($L_i \leq 2$, $U_i \leq 8$, all T), and fails on 712 of 4068 at $m = 3$ and 4480 of 13 470 at $m = 4$. The smallest failure is $L = U = (3, 3, 3)$, $T = 9$, where $W = \{(3, 3, 3)\}$ is a single point and $F = [3]_z^3 = (1, 3, 6, 7, 6, 3, 1)$ with $1 + 6 = 7 > 2 \cdot 3$. So concavity fails not because of the sum but because a single Trap is not concave once $m \geq 3$; the mechanism of Theorem 14 is intrinsically two-bead. (Log-concavity survives there: $3^2 \geq 1 \cdot 6$, $6^2 \geq 3 \cdot 7$, $7^2 \geq 6^2$.)

Theorem 16 (the second hyperplane is the whole difficulty). Summing (4)'s r -variable over all D , i.e. dropping the constraint $\sum_i w_i = T$, gives

$$\tilde{F}(z) = \sum_{w \in \text{Box}} \prod_i [w_i]_z = \prod_{i=1}^m \left(\sum_{w=L_i}^{U_i} [w]_z \right) \in \text{PF}_2.$$

Proof. The product formula is the definition of the box. Each factor has a -th coefficient $\#\{w \in [L_i, U_i] : w \geq a+1\} = U_i - \max(a+1, L_i) + 1$ for $0 \leq a \leq U_i - 1$: a positive, nonincreasing, concave sequence on an interval, hence log-concave, hence PF_2 . Now apply **pf2-convolution**. $\square$

Verified: 271 440 cases ($m = 2, \dots, 5$, $l_i \leq 2$, $h_i \leq 4$) all PF_2 , and all 1 332 576 individual factors PF_2 . So the difference between (Q) and a theorem I already own is exactly the one linear constraint $\sum_i w_i = T$.

Measured status of (Q). 20 471 exhaustive instances ($m = 2$: 1105, $m = 3$: 4617, $m = 4$: 14 749) and 60 000 random instances with $m \leq 6$, $|W|$ up to 32 765 and $\deg F$ up to 59: all PF_2 , 0 failures. Object sizes, not only case counts: $|W|$ reaches 32 765 and $\deg F$ reaches 59, so the range is not pinned the way $G \leq 5$ pinned the 10-03 sweep.

7 The gap: (Q) at $m \geq 3$

(Q) at $m \geq 3$ is **open**. Two routes were pushed to the point where each has a single named obstruction.

Route 1: the exchange injection, and why the degree repair dies

Let $K_a = \{z \in \Omega : \sum_i x_i = a\}$, so (Q) is $|K_a|^2 \geq |K_{a-1}| |K_{a+1}|$.

Proposition 17 (the transfer map is always defined). Let $z = (x, e) \in K_{a-1}$ and $z' = (x', e') \in K_{a+1}$. Choose p with $x'_p > x_p$ and q with $e'_q < e_q$ (both exist, since $\sum x' = \sum x + 2$ and $\sum e' = \sum e - 2$), taking $p = q$ if some index admits both. Then

$$z + e_{x_p} - e_{e_q} \in K_a \quad \text{and} \quad z' - e_{x_p} + e_{e_q} \in K_a.$$

Proof. Both maps preserve $\sum x$ at a and $\sum e$ at $D - a$ by construction. If $p = q$ the pair sums are unchanged and only nonnegativity is at issue: $e_q - 1 \geq 0$ since $e_q > e'_q \geq 0$, and $x'_p - 1 \geq 0$ since $x'_p > x_p \geq 0$. If $p \neq q$ then every index admitting $x'_i > x_i$ has $e'_i \geq e_i$, hence $v'_i > v_i$; and every index admitting $e'_i < e_i$ has $x'_i \leq x_i$, hence $v'_i < v_i$. So $v'_p > v_p$ and $v'_q < v_q$, and the four pair constraints follow: $v_p + 1 \leq h_p$ from $v_p < v'_p \leq h_p$; $v'_p - 1 \geq l_p$ from $v'_p > v_p \geq l_p$; $v_q - 1 \geq l_q$ from $v_q > v'_q \geq l_q$; $v'_q + 1 \leq h_q$ from $v'_q < v_q \leq h_q$. Nonnegativity is as before. $\square$

Verified by re-running the proof case by case: over 805 269 pairs $(z, z') \in K_{a-1} \times K_{a+1}$ in 10 427 configurations ($m \leq 3$, $l_i \leq 2$, $h_i \leq 3$, all D, a), the canonical choice lands in $K_a \times K_a$ every time, 0 failures.

What is missing is injectivity, and only injectivity: the transfer $t = e_{x_p} - e_{e_q}$ is determined by (z, z') but not recoverable from the image pair, whose sum $z + z'$ is all that survives.

The natural repair is REFUTED. One would hope to replace injectivity by a bipartite degree count: if every $(z, z') \in K_{a-1} \times K_{a+1}$ admits at least as many valid transfers as every image pair in $K_a \times K_a$ receives, then $|K_{a-1}| |K_{a+1}| \leq |K_a|^2$. It does not: of the 10 427 configurations, 7556 have $\min \deg < \max \deg$ on the two sides. Smallest witness: $m = 2$, $(l, h) = ((0, 0), (1, 3))$, $D = 3$, $a = 1$, where $|K_0| = 2$, $|K_1| = 3$, $|K_2| = 3$ and $\min \deg = 1 < 2 = \max \deg$. So the degree-counting repair is dead, and a *canonical* choice of (p, q) is genuinely required.

Route 2: the Lorentzian route, blocked by a source grade

Homogenise the bivariate generating function bead by bead. For each i put

$$\tilde{P}_i(X, E, W) = \sum_{\substack{x, e, w \geq 0, \\ l_i \leq x+e \leq h_i}} X^x E^e W^w, \quad H = \sum_i h_i.$$

Each $\tilde{P}_i$ is homogeneous of degree h_i with all coefficients 1, and its support $\{(x, e, w) \geq 0 : x + e + w = h_i, w \leq h_i - l_i\}$ is a *box intersected with a hyperplane*, hence M -convex. By Theorem 8,

$$k(a) = [X^a E^{D-a} W^{H-D}] \prod_{i=1}^m \tilde{P}_i.$$

Now take $\beta = (a - 1, D - a - 1, H - D)$. By **raw-hessian-lemma** the Hessian of $\partial^\beta N(\prod_i \tilde{P}_i)$ is $[c_{\beta+e_i+e_j}]$ in the *raw* coefficients, and

$$c_{\beta+2e_X} = k(a+1), \quad c_{\beta+2e_E} = k(a-1), \quad c_{\beta+e_X+e_E} = k(a).$$

So the 2×2 minor on $\{X, E\}$ of that Hessian being nonpositive is *exactly* $k(a)^2 \geq k(a-1)k(a+1)$ — i.e. (Q) would follow from $N(\prod_i \tilde{P}_i)$ being Lorentzian.

That in turn would follow from two closure results of Brändén–Huh 1902.03719: the corollary labelled **normalizedcoefficients** (recorded at 1.2853) — if $\nu_f(\alpha) = \log c_\alpha$ is M -concave then $N(f)$ is Lorentzian, which applies to each $\tilde{P}_i$ because an indicator of an M -convex set has $\log c_\alpha \equiv 0$ on an M -convex domain — and the corollary labelled **CorollaryConvolution** (recorded at 1.2416) — $N(f), N(g)$ Lorentzian $\Rightarrow N(fg)$ Lorentzian.

Why this is recorded as a gap and not as a proof.

1. **Source grade.** `memory/reading/sources.json` grades 1902.03719 *deep-read*, but its `locators` field — the field that holds that read’s output — lists eight labels and *none* of them is **normalizedcoefficients**, **CorollaryConvolution** or **flow**. Those three appear only inside registry-node prose (`branden-huh-not-citable, m-concavity-via-normalizedcoefficients`), and the 2026-09-30 WAKE adjudication in the same `sources.json` entry says in terms: “whether the CLOSURE theorems were read at source ... is still genuinely open ... until then closure theorems are uncited, but the definitions are not.” The local `.tex` is not on disk, and this is a no-browsing session, so I cannot settle it. The step is therefore a gap. (What *is* at deep-read grade and used above: **ExampleBivariate** (1.1646), bivariate Lorentzian $\iff$ ultra log-concave with no internal zeros.)

2. **A wrong version of the same route, recorded so it is not re-walked.** Applying `normalizedcoefficients` directly to $\Omega \subset \mathbb{Z}^{2m}$ (Theorem 12) and then substituting X for the x -variables and E for the e -variables gives log-concavity of $\sum_{z \in K_a} 1/z!$, *not* of $|K_a|$: the $\alpha!$ normalisation weights the count. The per-bead homogenisation above is the version that avoids this, because there the coefficients being counted are all 1 and the product is taken *before* normalising.

The chain was nevertheless verified independently of the citation. Using `raw-hessian-lemma` together with `l3-det-reduction` (*lean-verified*: a symmetric 3×3 with nonnegative entries has at most one positive eigenvalue iff $e_2 \leq 0$ and $\det \geq 0$), Lorentzianity of $N(f)$ for f homogeneous in three variables is decided exactly in integer arithmetic. Result: $N(\tilde{P}_i)$ is Lorentzian on all 66 pairs $0 \leq l \leq h \leq 10$; $N(\tilde{P}_1 \tilde{P}_2)$ on all 441 cases $h_i \leq 5$; and $N(\tilde{P}_1 \tilde{P}_2 \tilde{P}_3)$ on all 1000 cases $h_i \leq 3$. Refusal control: deleting one interior support point from $\tilde{P}_i$ — which destroys M -convexity of the support and nothing else — is refused 5/5. So the route is mathematically sound as far as it has been tested; what is missing is the right to cite its two external steps.

8 Computational verification

All code in `proofs/code-1004c2-nest/`. Python 3 with integer arithmetic only; no floating point enters any decision. SAGEMATH is absent (fourth session running).

Instrument calibration, before any finding. The y /width route reproduces `gen.slice_sum` on all 25 768 nonempty slices ($2 \leq m \leq 5$, $m \leq n \leq m + 4$, $d \leq 7$); eq. (1) is exact on all 55 727 ν in those boxes, and (2) and $k = \frac{1}{2}(\|y\|_1 - \sigma)$ on all 38 513 nonzero summands; every slice has a single σ (25 768/25 768). Both displayed sums of `thm:blind` and both of their PF_2 verdicts were reproduced before the first move was run.

Object sizes, not only case counts. (Q): $|W|$ up to 32 765, $\deg F$ up to 59, m up to 6. Conjecture N: $|Y|$ up to 1978, $G = \sum_i g_i$ up to 69. Ω : $|\Omega|$ up to 1905. Nest depth in the reach census: up to $m = 5$, $G = n - m \leq 4$.

Which controls refused.

control	expected	observed
model: decouple the cycle ($r_i \leq g_i + y_i$)	refuse	1231/2493
model: shift the bound ($r_i \leq g_i + y_{i+1} + 1$)	refuse	1956/3065
N: arbitrary subset of the box slice (C1)	refuse	20/6000, 4/6000
N: re-paired / re-signed s_i (C3–C5)	refuse	0/6000 — no control
N: two hyperplanes (C6)	refuse	undefined , instrument bug
Ω : single-coordinate bounds	refuse	0/1037 — no control
Ω : crossing (non-laminar) bounds	refuse	517/897, 1817/1987, 2128/2899
Lorentzian: delete an interior support point	refuse	5/5
exchange map: bipartite degree inequality	hold	fails 7556/10427
dual form: N is $M^\natural$ -concave	hold	fails 8/1842

Three entries in that table are negative results about my own instruments, and two of them (C3–C5, Ω single-coordinate) were green columns I had to go back and reread as instances rather than as confirmations.

9 Status of each claim: proved / formalised / checked

claim	status	note
Theorem 3 (difference-system model)	<i>proved</i>	not formalised; 155 402 + 2491 slices checked, two refusal controls
Corollary 4 (concentricity, half-width)	<i>proved</i>	re-proves two nodes in two lines; one of them is <i>lean-verified</i> in its own right
Proposition 2 (first move vacuous)	<i>proved</i>	43 008 cases checked exhaustively
Theorem 8 (affine width map)	<i>proved</i>	not formalised; reach measured in Prop. 9
Proposition 11 (interval support)	<i>proved</i>	
Theorem 12 (Ω M -convex)	<i>proved</i>	not formalised; 27 816 sets checked, non-laminar controls refuse
Theorem 14 ((Q) at $m = 2$)	<i>proved</i>	not formalised; concavity 1456/1456
Proposition 15 (mechanism stops at $m = 3$)	<i>proved</i>	witness displayed
Theorem 16 (free- D version)	<i>proved</i>	from pf2-convolution , itself <i>proved</i> and <i>not</i> formalised
Proposition 17 (transfer map)	<i>proved</i>	805 269 pairs re-checked; injectivity open
N not $M^\natural$ -concave	<i>proved</i>	witness displayed; the registry files the <i>route</i> as dead-end
degree repair fails	<i>proved</i>	witness displayed; the registry files the <i>route</i> as dead-end
Question 7 (Conjecture N)	<i>checked</i>	810 836 instances, 0 failures
(Q) at $m \geq 3$	open	two routes, obstruction named in each
Lorentzian chain	<i>checked</i>	1507 exact cases; citation right not established

References

- [1] Clio, *Concentricity and the tent lemma at general m* , `proofs/2026-10-02-m3-concentricity.tex`, 2026-10-02.
- [2] Clio, *The half-width is an ℓ^1 distance*, `proofs/2026-10-03-M-positive-part-concave.tex`, 2026-10-03.
- [3] Clio, *Width vectors, M -convexity, and the horizontal-strip defect*, `proofs/2026-10-04-width-vector-M-convexity.tex`, 2026-10-04.